



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING) EXAMINATIONS - JANUARY/FEBRUARY 2023

SUBJECT : CSE 2151 - COMPUTER ORGANIZATION AND ARCHITECTURE (MAKEUP)

COMPUTER ORGANIZATION AND ARCHITECTURE [CSE 2151]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Assume that an array A is stored in consecutive word locations starting from address A and an array B is stored in consecutive word locations starting from address B. Write a CISC-style program to find the product of the corresponding elements of two arrays and store it in the third array C starting at address C. Assume uniform sized arrays with the size of the array stored in memory location SIZE. Assume byte addressable memory and word length is two bytes. (3)
- 1B) Represent the following integers in 2's complement representation with minimum number of bits required, perform the operation mentioned and state whether overflow occurs or not in each case. Also, if overflow has occurred, state why it has occurred. (3)
- i) Add -8 with -7
- ii) Subtract +15 from -16
- 1C) Consider two decimal numbers $A = +81.375$ $B = +15.5$
Perform the following operations: (4)
- a) Convert the decimal numbers to floating point binary
- b) Normalize the number obtained above in (a)
- c) Represent the normalized numbers in IEEE32 bit floating point representation
- d) Perform (A / B) , on the numbers represented in (c) using the rules for division of IEEE 32 bit floating point representation. Indicate all the steps clearly
- e) Store the result in the IEEE 32-bit floating-point representation
- 2A) Considering 1010 as Dividend and 0011 as Divisor, perform the division using restoring division algorithm. (3)
- 2B) Depict the Hardware Implementation of Multiplication of unsigned Binary Numbers. (2)
- 2C) In context of the hardwired control unit design, draw the state diagram of the Booth's Multiplier Controller along with the steps in its controller action. In what ways, the Hardwired Control Design Mechanism is disadvantageous in comparison with that of the Microprogrammed Control Unit Design Mechanisms. (5)
- 3A) Design a Microprogrammed control unit using Wilke's design for the Register Transfer Description given below: (3)

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    Declare registers A[8], B[8]
    Declare inbus[8], outbus[8];
Start:  A ← inbus, B ← 0;
Back:  If A[1:0]=00 then go to Add;
       go to Next;
Add:   B ← B+A;
Next:  A ← A-1;
       If A>0 then go to Back;
       Outbus ← B;
Halt:  go to Halt

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- 3B) A computer has a small data cache capable of holding eight words and has a byte addressable memory. Each cache block consists of one word and one word comprises of one byte. When a given program is executed, the processor reads data sequentially from the addresses given below (Addresses are given in decimal). Assume that the cache is initially empty. Show the contents of the cache after each memory access, for a Direct mapped cache, and compute the hit rate (5)
7200, 7201, 7208, 7213, 7200, 7216, 7208, 7201, 7215, 7201, 7213, 7218, 7220, 7213, 7217, 7219
- 3C) A disk unit has 25 recording surfaces. It has a total of 16,000 cylinders. There is an average of 200 sectors per track. Each sector contains 512 bytes of data. What is the maximum number of bytes that can be stored in this unit? What is the data transfer rate in bytes per second at a rotational speed of 7200 rpm? (2)
- 4A) Design a large memory consisting of 4G words of 64 bits each for a system. There are 2 types of static memory chips available of same physical size, $512M \times 8$ and $1024M \times 16$. Identify the most suitable memory chip such that a smaller decoder is used and justify the answer. Draw the structure and indicate clearly, the number of address lines and the data lines for large memory chips in the diagram. (3)
- 4B) A computer has a data cache capable of holding 16 words and each word is one byte. The cache block consists of two words and initially the cache is empty. When the program is executed, the processor reads the data from the following decimal addresses given below. The data is stored in continuous location of the main memory starting from location 1000 to 1031. Show the organization of the words in the main memory. Compare and identify the most suitable mapping technique to solve this problem. Show the contents of cache using each technique. If blocks are grouped into sets, then a maximum of 2 blocks can be grouped into a single set. (5)
1000, 1003, 1004, 1007, 1009, 1010, 1012, 1015
- 4C) Distinguish between UMA and NUMA multiprocessor as far as network latency is concerned. (2)
- 5A) The Informatics Business unit of Delta Information Solutions, a Multinational Company operates on shared memory multiprocessors which demands an interconnection that requires a direct link between any pair of units connected to the network to connect its primary constituent units namely the FinTech Group, the Technology Solutions Group, Business Solutions Group, Technology Analytics Group and Market Research Group. Assume, each of these groups is an independent processing unit. Identify the type of interconnection network and explain its structure diagrammatically. Discuss the Ring Based Interconnection Networks along with the hierarchy of rings. (5)
- 5B) Give the diagrammatic depiction of the five-stage pipelining architecture and enlist the significance of the various interstage buffers in the architecture drawn. (3)

5C) Distinguish between Fine-Grained Multithreading and Coarse-Grained Multithreading. (2)

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